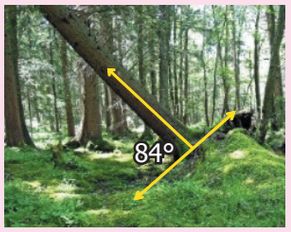
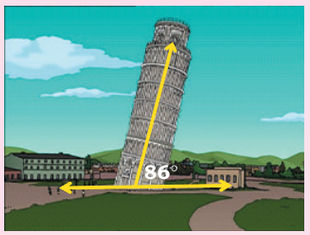
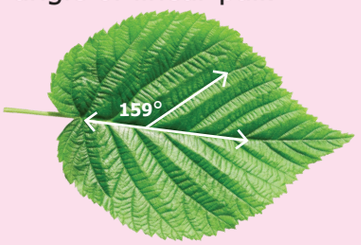
Try these

1. Observe the following pictures and find the other angle of linear pair.

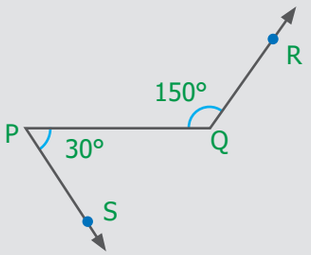






Think

Observe the figure. There are two angles namely $\angle PQR = 150°$ and $\angle QPS = 30°$. Is all this pair of supplementary angles a linear pair? Discuss.



Example 5.1 In Fig. 5.5, find $\angle AOC$.

Solution

$$\angle AOC = \angle AOB + \angle BOC$$
$$= 51° + 46°$$
$$= 97°$$

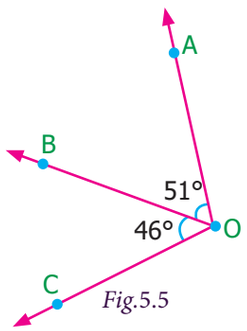


Fig.5.5

Example 5.2 If $\angle POQ = 23°$ and $\angle POR = 62°$ then find $\angle QOR$

Solution

We know that $\angle POR = \angle POQ + \angle QOR$

$62° = 23° + \angle QOR$

Subtracting 23° on both sides

$62° - 23° = 23° + \angle QOR - 23°$

$\angle QOR = 39°$

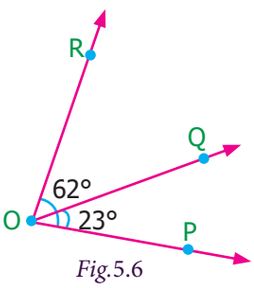


Fig.5.6

Example 5.3 Which of the following pair of adjacent angles will make a linear pair?

(i) 89°, 91° (ii) 105°, 65° (iii) 117°, 62° (iv) 40°, 140°

Solution

(i) Since 89° + 91° = 180°, this pair will be a linear pair.

(ii) Since 105° + 65° = 170° ≠ 180°, this pair cannot make a linear pair.

(iii) Since 117° + 62° = 179° ≠ 180°, this pair cannot make a linear pair.

(iv) Since 40° + 140° = 180°, this pair will be a linear pair.

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